

Lab - Guideline

Criteria to consider when choosing network media (from CISCO CCNA1).

What is the maximum distance that the media can successfully carry a signal?

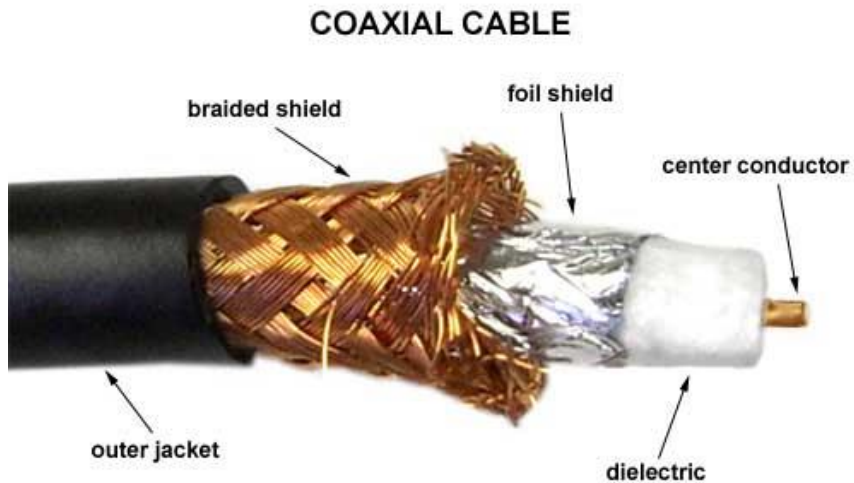
Into what type of environment will the media be installed?

What is the amount of data and the speed at which it must be transmitted?

What is the cost of the media and installation?

A. Coaxial cables

- 10BASE2
- 10BASE5



B. Twisted cables

- UTP
- FTP
- STP

Cat5e: 10/100/1000 Mbps over a max of 100m distance

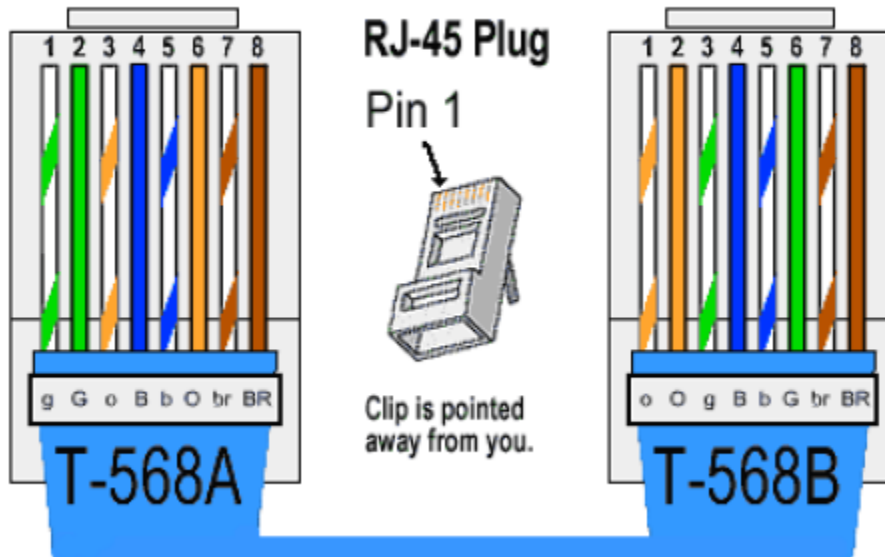
Cat6: 1Gbps over a max of 100m distance and 10Gbps over 55m

Cat6a: 1/10Gbps over a max of 100m distance

RJ45 connector (part of 8P8C modular connectors), Cable Crimper Plier, punch down tool

EIA/TIA-T568-A and EIA/TIA-T568-B

RJ-45 Crossover Ethernet Cable



Patch cord, Patch panel, Hub = multiport repeater
Cable Tester

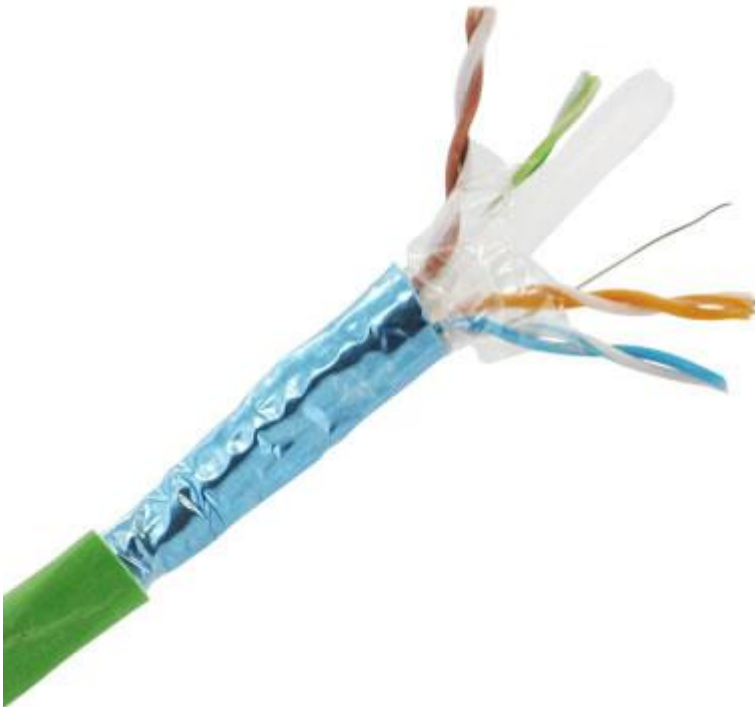


	Category 5e Class D	Category 6 Class E	Category 6A Class E _A	Category 7 Class F	Category 7 _A Class F _A
Bandwidth	100 MHz	250 MHz	500 MHz	600 MHz	1,000 MHz
Cable type	UTP	UTP/FTP	UTP/FTP	S/FTP	S/FTP
Insertion loss (dB)	24	21.3	20.9	20.8	20.3
NEXT loss (dB)	30.1	39.9	39.9	62.9	65
ACR (dB)	6.1	18.6	19	42.1	44.1

UTP = Unshielded twisted pair

FTP = Foil twisted pair

S/FTP = Shielded/foil twisted pair



cat6a

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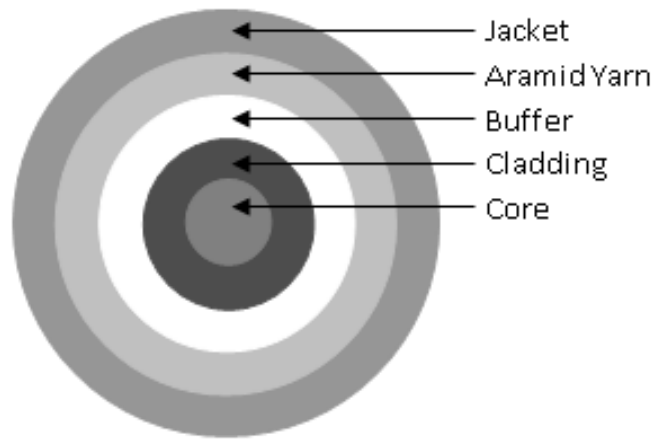
What is the amount of data and the speed at which it must be transmitted?

What is the cost of the media and installation?

www.submarinecablemap.com

A. FO

- Phenomenon: *total internal reflection*,
- Optical communication systems use high carrier frequencies (~ 100 THz) in the visible or near-infrared region of the electromagnetic spectrum.
- telecommunication applications: short-haul (intra-city, usually < 50 km) and long-haul (inter-city)



- multimode fiber
 - 62.5/125 μ or 50/125 μ .
 - 100Mbps $\rightarrow$ 2000m; 1 Gbps up to 1000m; 10Gbps up to 550 m
 - loss: 1-2dB/km
- single-mode fiber
 - 9/125 μ
 - up to 10 Gbps over 80 km (50 mi); DWDM: thousands of kilometers at 10 Gbps; hundred kilometers at 40 Gbps
 - loss: 0,5dB/km
- optical transmitter
 - LED, LASER on different wavelength
- optical receiver
 - photo-diodes
- junctions: mechanical and fusion, with loss of 0.5dB and 0.1dB
- connectors: SC and ST with loss of 1dB
- repeaters
- patch-panels
- budget of loss of optical link
- Testing
 - power output
 - *bit error rate* (BER) = average probability of incorrect bit identification; values: 10^{-9} - 10^{-14}

- OTDR Optical Time Domain Reflectometer

- *optical amplification* for increasing the repeater spacing and of *wavelength-division multiplexing* (WDM) for increasing the bit rate
- amplifiers spaced at 60-70 km apart
- Capacity:
 - **40 and 100Gb Ethernet**
 - commercial WDM systems exceeded **1.6 Tb/s**.
 - commercial WDM operated by VSNL: **2.56 Tb/s** (64 WDM channels at 10 Gb/s over 4 fiber pairs)
 - world record: **1125 terabits per second**, University College London (UCL)

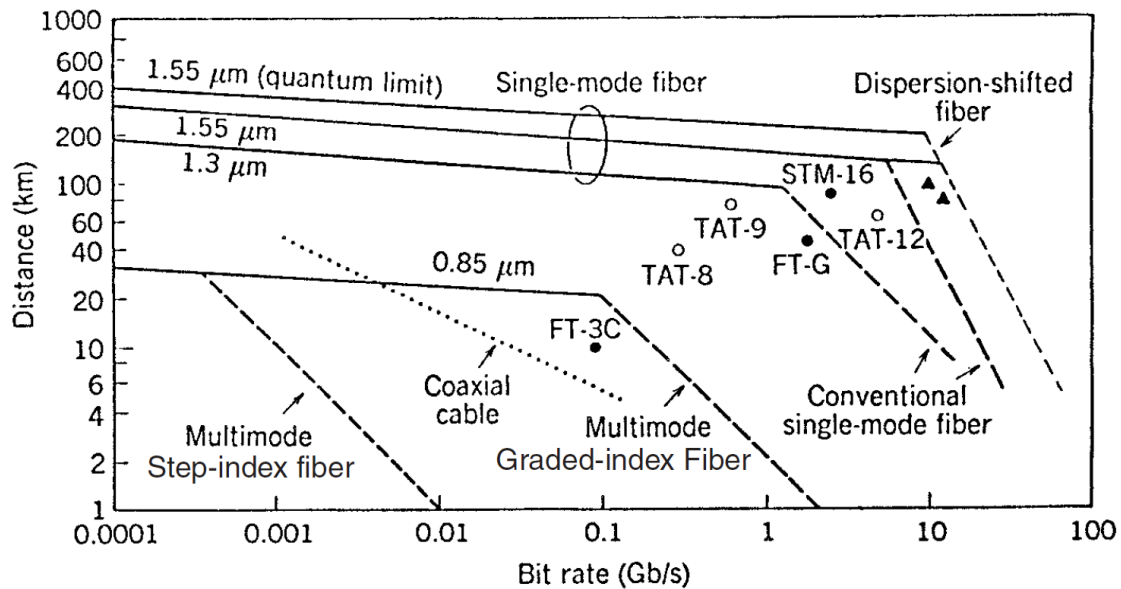


Figure 5.4: Loss (solid lines) and dispersion (dashed lines) limits on transmission distance L as a function of bit rate B for the three wavelength windows. The dotted line corresponds to coaxial cables. Circles denote commercial lightwave systems; triangles show laboratory experiments. (After Ref. [1]; ©1988 Academic Press; reprinted with permission.)

3.2 A 9/125 μ single-mode optical fiber having the length of 2,5km and the loss equal to 0,5dB/km, which connects two DTE equipments is considered. The attenuation introduced by junctions and connectors is equal to 0,5 and 1dB respectively. The error margin taken into consideration is 3dB. The power of average emission of the transmitter is -15dB, the receiver sensitivity at a rate of errors given by BER 10^{-9} is -25dB and dynamic of the receiver is in the range -10 ÷ -30dB. Calculate the optical power budget.

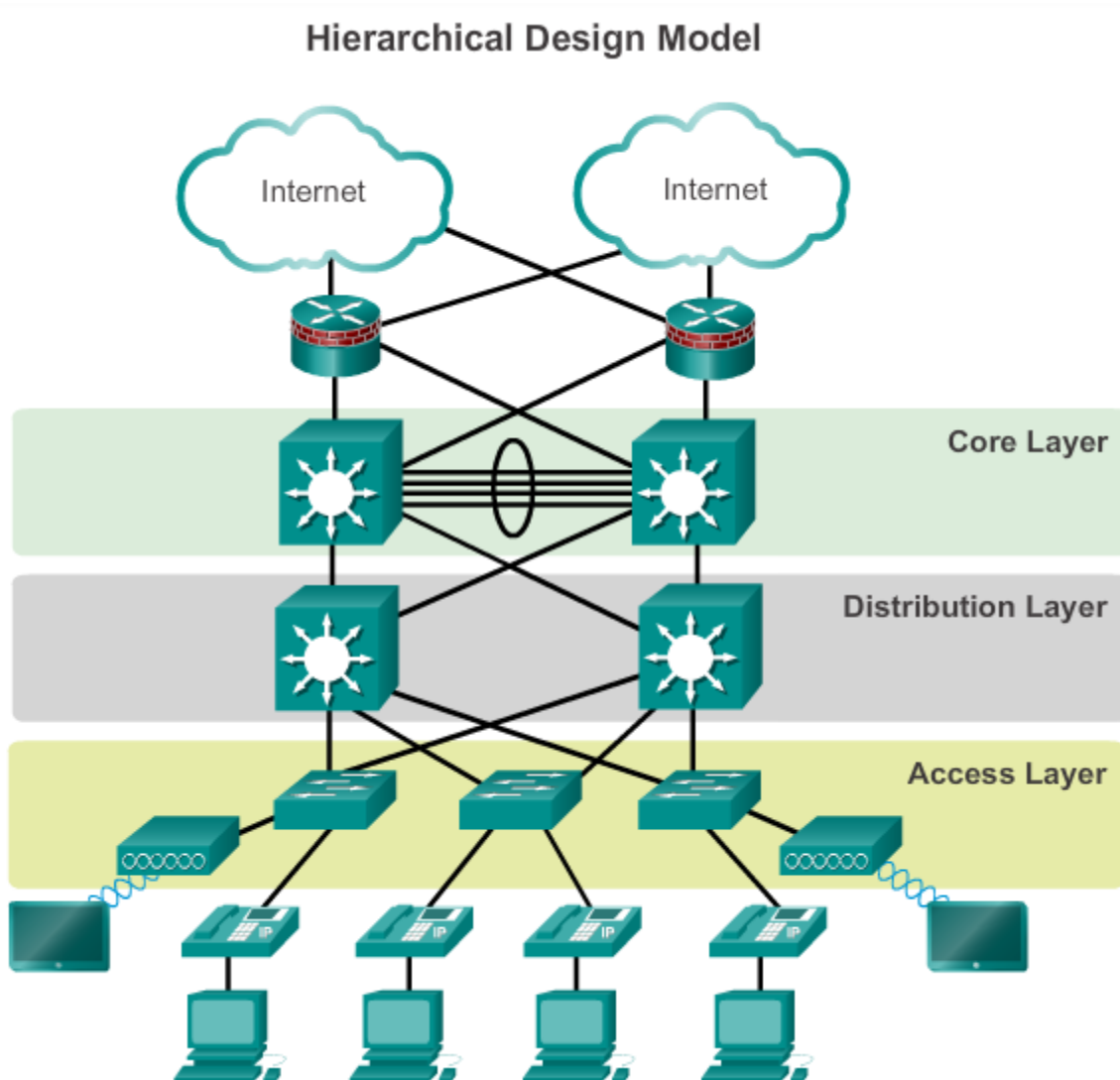
Crt.	Optical loss or power	DB
1.	The km loss in Optical Fibre db/km X _____ km fibre	_____ dB
2.	The loss in Junctions _____ dB/junction X _____ junctions	_____ dB
3.	The loss in Connectors _____ dB/connector X _____ connectors	_____ dB
4.	Losses on other components	_____ dB
5.	Margin of error	_____ dB
6.	Total loss on the Link (1+2+3+4+5)	_____ dB
7.	The power of average emission of the transmitter	_____ dB
8.	Average power received by the receiver (7-6)	_____ dB
9.	The dynamic of the receiver _____ dB at _____ dB	
10.	Receiver sensitivity at a rate of errors given by BER	_____ dB
11.	Available Remaining Power	_____ dB

B. Structured Cabling

Hierarchical Network Design

Three-tier model:

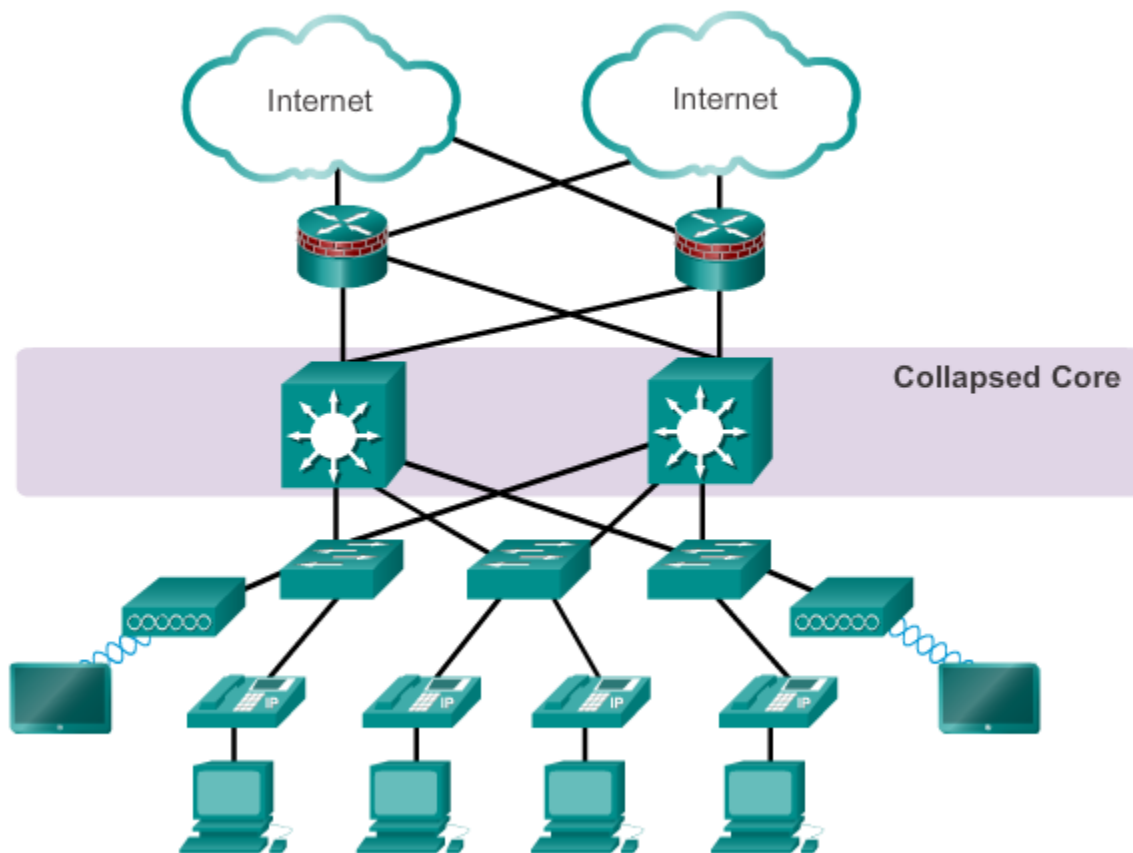
- enterprise networks
- core layer: high-speed backbone layer between dispersed networks



Two-tier model:

- smaller enterprise networks
- core and distribution layers are collapsed into one layer

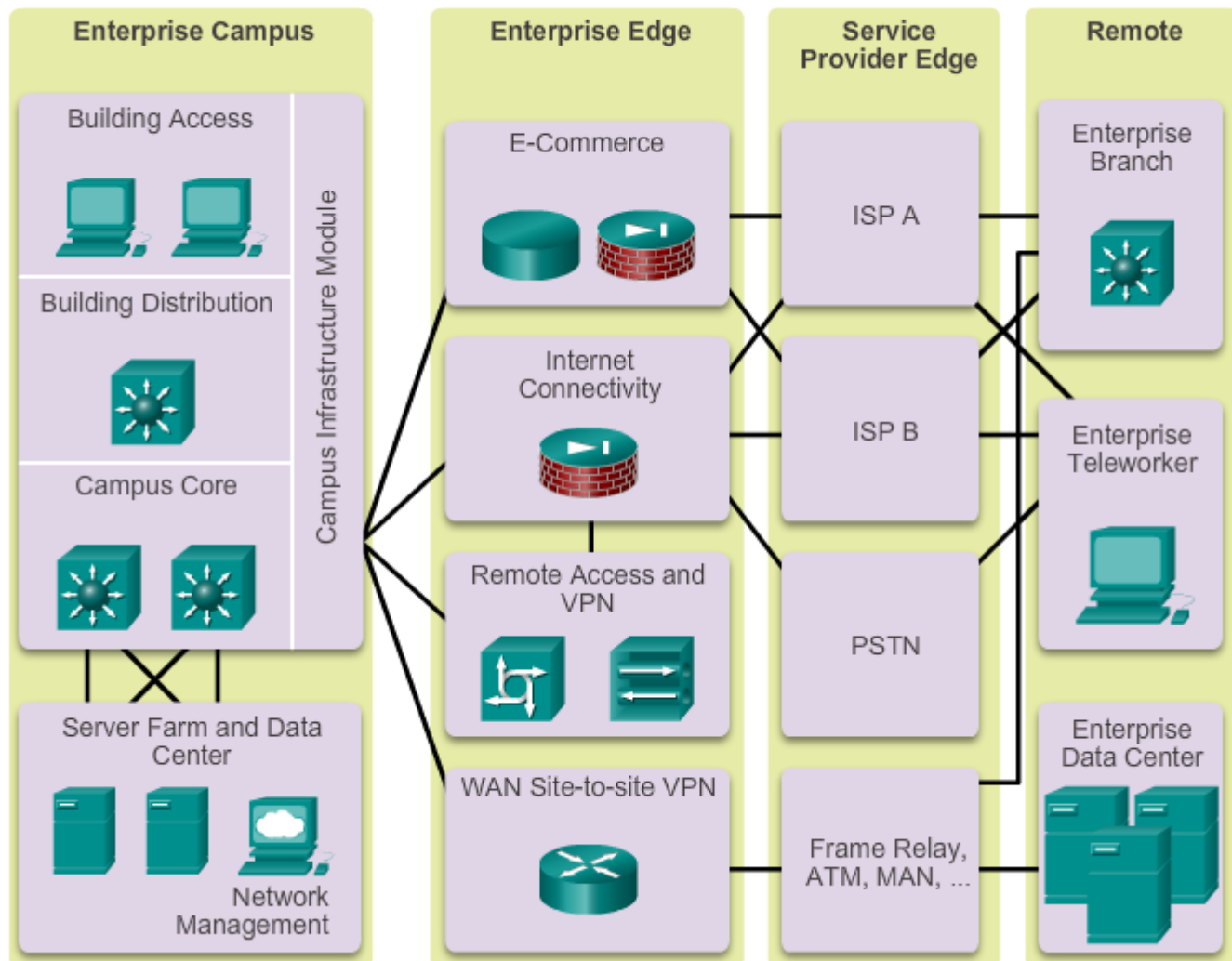
Collapsed Core



Cisco Enterprise Architecture

- core layer module-> high-speed interconnectivity between:
 - distribution layer modules
 - data center server farms
 - the enterprise edge

Enterprise Architecture



Choosing devices

Common Business Considerations When Selecting Switch Equipment:

- **Cost** - The cost of a switch will depend on the number and speed of the interfaces, supported features, and expansion capability.
- **Port Density** - Network switches must support the appropriate number of devices on the network.
- **Power** - It is now common to power access points, IP phones, and even compact switches using Power over Ethernet (PoE). In addition to PoE considerations, some chassis-based switches support redundant power supplies.
- **Reliability** - The switch should provide continuous access to the network.
- **Port Speed** - The speed of the network connection is of primary concern to end users.
- **Frame Buffers** - The ability of the switch to store frames is important in a network where there may be congested ports to servers or other areas of the network.
- **Scalability** - The number of users on a network typically grows over time; therefore, the switch should provide the opportunity for growth.

- a. Fixed Configuration Switches
- b. Modular Configuration Switches
- c. Stackable Configuration Switches: Cisco StackWise technology allows the interconnection of up to nine switches (fault tolerance and bandwidth availability)

Fixed Configuration Switches



Features and options are limited to those that originally come with the switch.

Modular Configuration Switches



The chassis accepts line cards that contain the ports.

Stackable Configuration Switches



Stackable switches, connected by a special cable, effectively operate as one large switch.

Switches:

1. Layer 2 switch
 2. Layer 3 switch
-
1. “form factor” switches
 2. modular switches
-
1. Switching speed: millions of packets per second
 2. Backplane speed: how fast traffic can be transmitted between modules in a switch
 3. Buffer size
 4. Management and monitoring

<http://www.enterprisenetworkingplanet.com/reports/index.php/28911/Enterprise+Ethernet+Switch+Buyers+Guide.htm>

Example:

Cisco Enterprise: <http://www.cisco.com/c/en/us/products/switches/index.html>

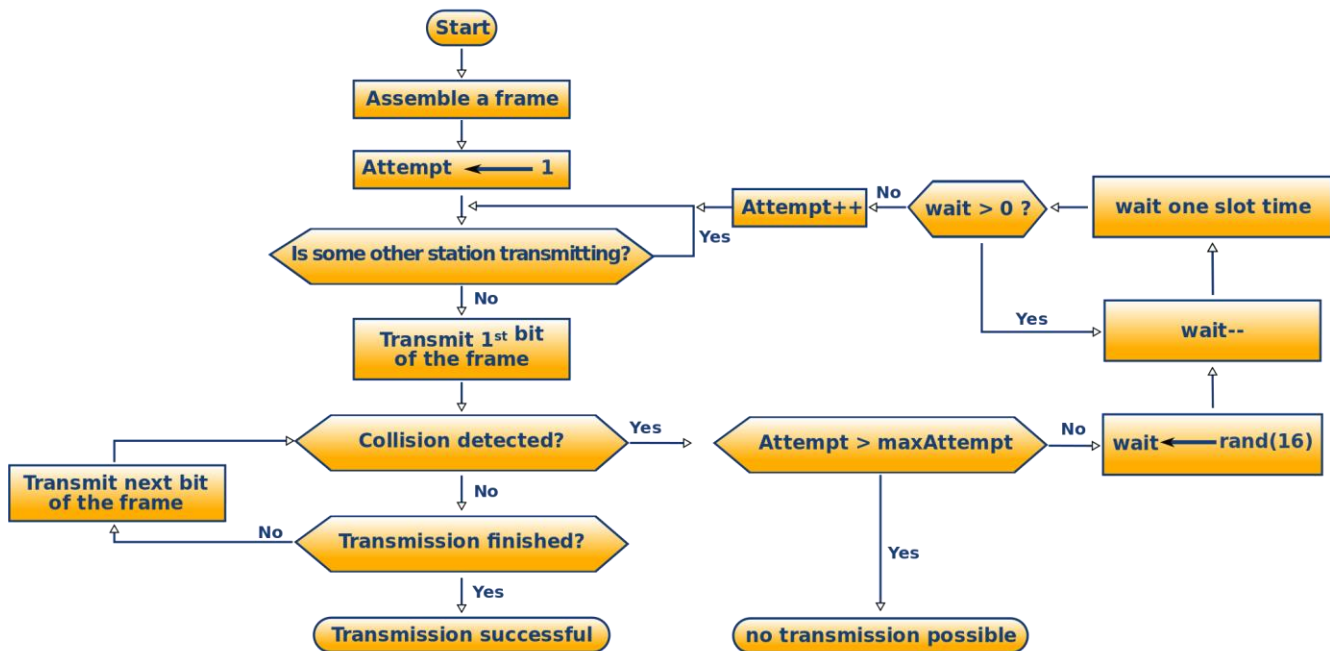
Lab - Guideline

Topics

- A. Metode de acces la mediu
- B. Cisco Packet Tracer

A. Metode de acces la mediu

- CSMA/CD



- Ethernet II

Ethernet II Frame Structure and Field Size					
8 Bytes	6 Bytes	6 Bytes	2 Bytes	46 - 1500 Bytes	4 Bytes
Preamble	Destination Address	Source Address	Type	Data	Frame Check Sequence

- IEEE 802.3 Ethernet
 - 10 Mbps – 10Gbps Ethernet networks, 40Gbps and 100Gbps Ethernet networks
 - Lungime minima frame (10/100Mbps): 512 biti (64 Bytes)
 - Lungime minima frame (1Gbps and higher): 4096 biti (512 Bytes)
 - Padding necesar!

$$T_{\text{jamming}} < T_{\text{cadru}}$$

- IEEE 802.4 – TOKEN-BUS
- IEEE 802.5 – TOKEN-RING

B. Introducere in utilizarea simulatorului Cisco Packet Tracer

Se va realiza o configurație minimală de bază pentru o rețea ce va consta din următoarele componente: două calculatoare legate printr-un switch de un router. La finalul lucrării de laborator toate dispozitivele trebuie să fie configurate și să poată comunica între ele (să răspundă la ping). Rețeaua este cea din figura 1:

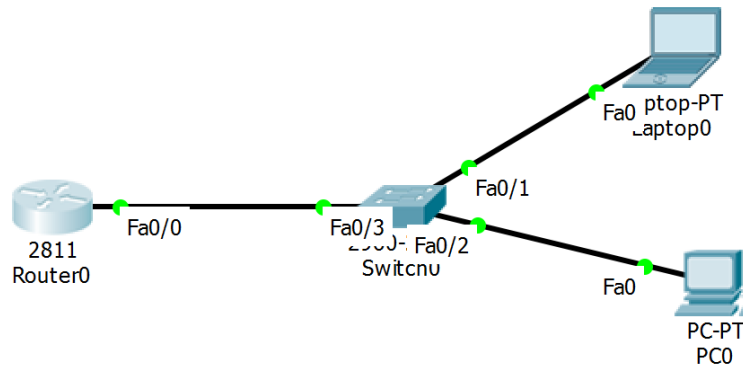


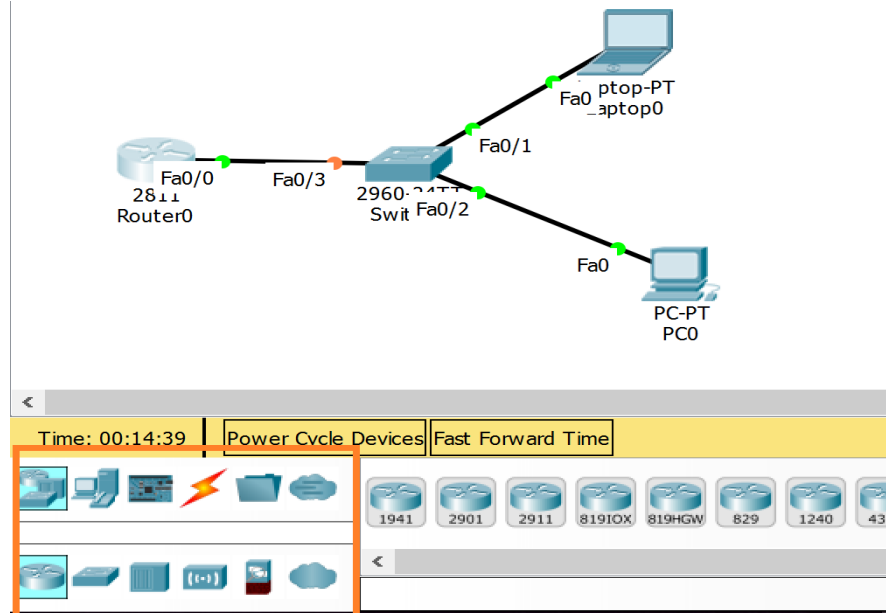
Figura 1. Configurația rețelei

Dispozitiv	Interfață	Adresă IP	Mască de rețea	Default gateway
Router0	Fa0/0	172.16.0.1	255.255.0.0	
PC0	NIC	172.16.0.2	255.255.0.0	172.16.0.1
Laptop0	NIC	172.16.0.3	255.255.0.0	172.16.0.1

Tabel 1 Adresele dispozitivelor

Se vor efectua următorii pași:

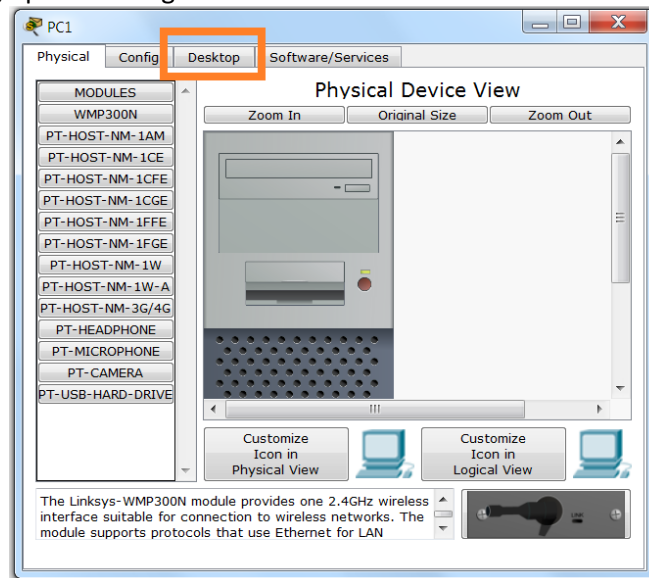
- se vor plasa în zona de lucru dispozitivele: cele două stații, un switch, un router și serverul (se selectează tipul echipamentului, de exemplu router, din fereastra din stânga jos, apoi modelul echipamentului din fereastra alăturată după care se dă click în spațiul de lucru în locul unde se dorește amplasarea sa). Se pot selecta anumite modele (care sunt gata "echipate" cu anumite interfețe), sau se pot folosi modele generice (caz în care interfețele switchurilor și routerelor vor trebui adăugate de utilizator);



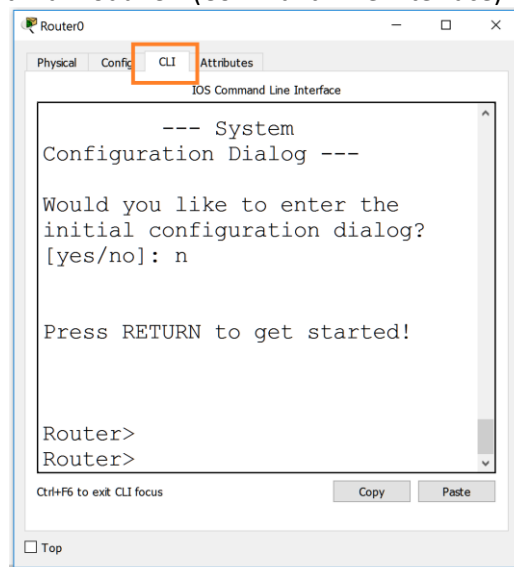
- se vor realiza conexiunile fizice ca în figură: cele două stații se vor lega cu un cablu drept la switch, routerul se va conecta cu switchul tot cu un cablu drept. Pentru realizarea unei conexiuni se va selecta din fereastra din stânga jos simbolul “Connections”, apoi din fereastra de lângă ea tipul conexiunii (cablu drept, inversor, serial etc.), după care se dă click pe fiecare dintre cele două dispozitive care se doresc a fi conectate, la fiecare dispozitiv alegându-se interfața la care se dorește realizarea conexiunii.



- se vor realiza configurările minimale ale stațiilor și ale routerului (adrese IP, măști de rețea, default gateway etc), conform tabelului de mai sus. La nivelul stațiilor se va da click pe fiecare stație, se va alege tabul Desktop, apoi IP Configuration.



- La nivelul routerului se va utiliza modul CLI (Command Line Interface):



- se va testa conectivitatea locală între cele două stații conectate în switch - acestea ar trebui să poată schimba pachete între ele (ping). Pentru aceasta se va selecta un calculator, apoi din tabul Desktop se va selecta “Command Line” și de acolo se va executa comanda ping.

Configurarea routerului presupune setarea următorilor parametrii:

- din linia de comandă se vor da următoarele comenzi pentru routerul 0 (pentru intrarea în modul linie de comandă se va da click pe router, apoi se selectează tabul CLI):
 - enable
 - configure terminal
 - interface Fa0/0
 - ip address 172.16.0.1 255.255.0.0
 - no shutdown
 - end
- Pentru verificarea interfețelor se poate folosi comanda `show interfaces` sau `show running-config`, la nivelul routerului.

După realizarea configurării de bază, de pe fiecare dispozitiv se va testa conectivitatea cu alte dispozitive direct conectate, de exemplu de pe router 1 va trebui să funcționeze comanda `ping` spre adresa stațiilor PC0 și Laptop0.

Pentru testarea conectivității se vor realiza următorii pași:

- se va deschide o linie de comandă pe stația PC0 și se va da `ping` la următoarele adrese:
 - adresa stației Laptop0 (172.16.0.3);
 - interfața Fastethernet0/0 de la routerul 0 (172.16.0.1);

Anexa1

Comenzi de baza in CLI:

- help

>?

- vizualizarea unor informatii despre comenzi/configuratii:

>**show <comanda>**

- accesarea modului privilegiat

>**enable**

- accesarea modului general de configurare

#configure terminal SAU #conf t

- accesarea modului de configurare a unei interfete de retea denumita "fastEthernet0/0"

#interface fastEthernet0/0

- accesarea modului de configurare a unei interfete VLAN denumita "vlan 10"

#interface vlan 10

- activarea unei interfete

#no shutdown

- dezactivarea unei interfete

#shutdown

- iesirea dintr-un mod de configurare

#exit SAU #end

- vizualizarea informatiilor IP ale interfetelor de retea (dupa iesirea din modul de configurare cu comanda **end**)

#show ip interface brief

- vizualizarea configuratiei echipamentului (dupa iesirea din modul de configurare cu comanda **end**)

#show running-config

- iesirea din modul privilegiat

#disable

Lab - Guideline

Topics

- A. Subnetting IPv4
- B. Router

A. Subnetting IPv4 (32 bits) – Internet Protocol version 4

IP address example:

Decimal notation	Binary Notation
74.125.43.99	01001010.01111101.00101011.01100011
	⏟ 8bits ⏟ 8bits ⏟ 8bits ⏟ 8bits

- *Static vs. dynamic addressing*
- *IPv4 address structure*

network ID	host ID
-------------------	----------------

Class	1 st Octet Decimal Range	1 st Octet High Order Bits	Network/Host ID (N=Network, H=Host)	Default Subnet Mask
A	1 – 126*	0	N.H.H.H	255.0.0.0
B	128 – 191	10	N.N.H.H	255.255.0.0
C	192 – 223	110	N.N.N.H	255.255.255.0
D	224 – 239	1110	Reserved for Multicasting	
E	240 – 254	1111	Experimental; used for research	

Note: Class A addresses 127.0.0.0 to 127.255.255.255 cannot be used and is reserved for loopback and diagnostic functions.

- *Network Mask (defaults)*

Class A: 255.0.0.0 or /8 (11111111.00000000.00000000.00000000) – first 8 bits are 1, in binary notation

Class B: 255.255.0.0 or /16 (11111111.11111111.00000000.00000000) – first 16 bits are 1, in binary notation

Class C: 255.255.255.0 or /24 (11111111.11111111.11111111.00000000) – first 24 bits are 1, in binary notation

- *Private IP addresses*

Class A: 10.0.0.0 - 10.255.255.255 /8
Class B: 172.16.0.0 - 172.31.255.255 /16
Class C: 192.168.0.0 - 192.168.255.255 /24

	172	25	114	250	AND	
IP Address (B class)	10101100	00011001	01110010	11111010		0 0 0
Network Mask	11111111	11111111	00000000	00000000		0 1 0
	255	255	0	0		1 0 0
Network Address	10101100	00011001	00000000	00000000	AND	1 1 1
	172	25	0	0		
Broadcast Address	10101100	00011001	11111111	11111111		
	172	25	255	255		

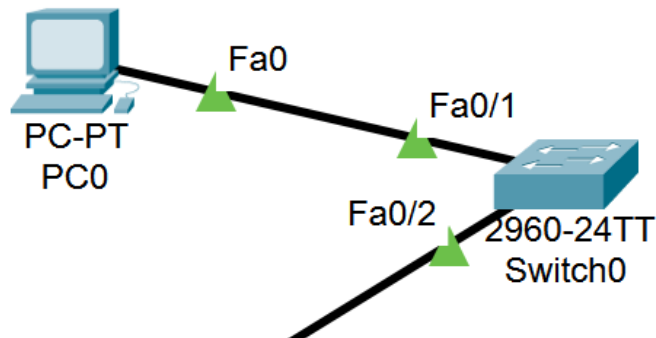
- Network address: all host bits = 0
- Network broadcast address: all host bits = 1
- Total number of host bits: $2^{16} = 65,536$
- Number of hosts: $2^{16} - 2 = 65,536 - 2 = 65,534$

!IMPORTANT: the **Network Address** and the **Broadcast Address** will **never be assigned** to a host (router/switch/computer etc)!

1. Our network IP address range is: 172.25.0.0 to 172.25.255.255 with the NM /16
2. The valid host range is: 172.25.0.1 to 172.25.255.254 with the NM /16
Remember: The first (Network Address) and last address (Broadcast Address) of this range can't be used for hosts

Example: a valid address to assign to the computer PC0 could be 172.25.34.222 /16 (a valid address from the host range)

IP: 172.25.34.222
 NM: 255.255.0.0



Exercise example (fill in the table):

IPv4 Address/Prefix	Network Address	Broadcast Address	Total Number of Host Bits	Total Number of Hosts
74.125.43.99 /26				

How to solve it:

IP: 74.125.43.99 = 01001010.01111101.00101011.01100011

NM: /26 (first 26 bits are 1, in binary notation) = 11111111.11111111.11111111.11000000

From the NM we learn:

- The first 26 bits in the IP address represent the Network ID (network bits)
- The last 6 bits in the IP address represent the Host ID (host bits)

IP logic AND with the NM:

01001010.01111101.00101011.01100011
 11111111.11111111.11111111.11000000

AND

01001010.01111101.00101011.01000000 – Network Address (all host bits are set to 0)
 74.125.43.64

01001010.01111101.00101011.01111111 – Broadcast Address (all host bits are set to 1)
 74.125.43.127

Question: which addresses could be assigned to a network device?

Our network IP address range is: 74.125.43.64 to 74.125.43.127. However, the first (Network Address) and last address (Broadcast Address) of this range can't be used.

Answer: any address, except the first and last address. So the valid host range is: 74.125.43.65 - 74.125.43.126

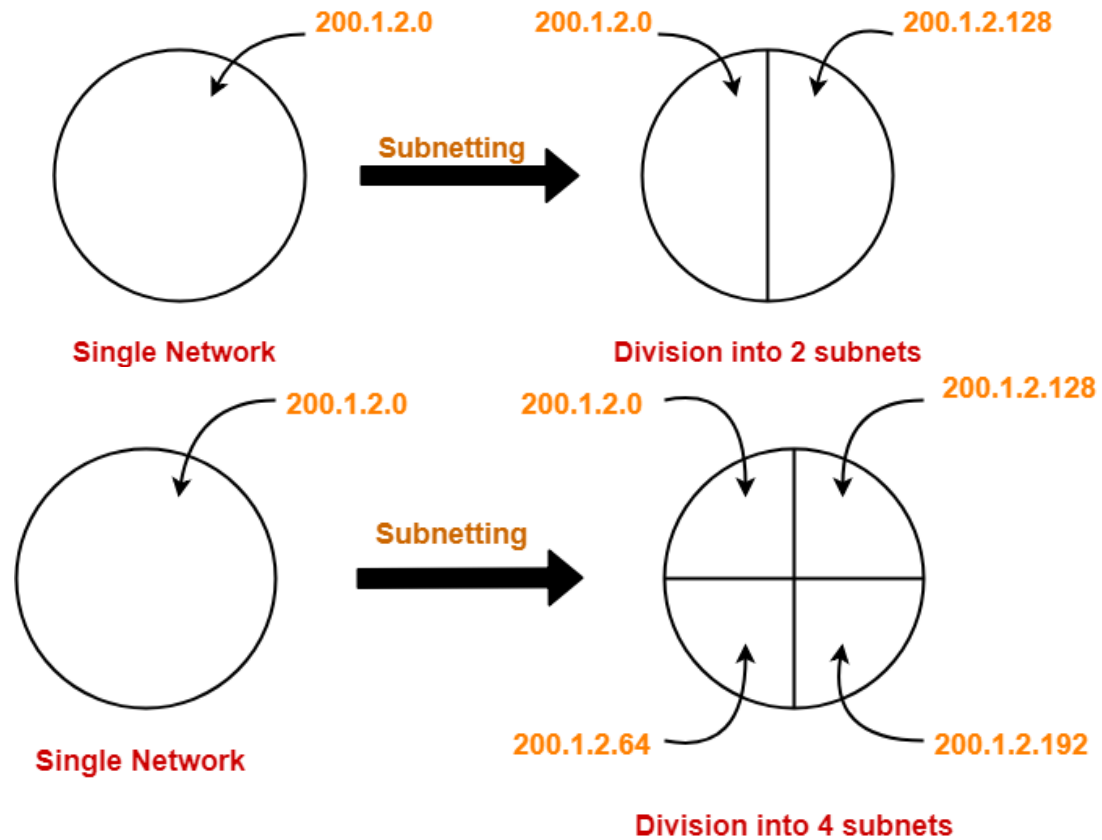
Result:

IPv4 Address/Prefix	Network Address	Broadcast Address	Total Number of Host Bits	Total Number of Hosts
74.125.43.99 /26	74.125.43.64	74.125.43.127	6	2 ⁶ -2

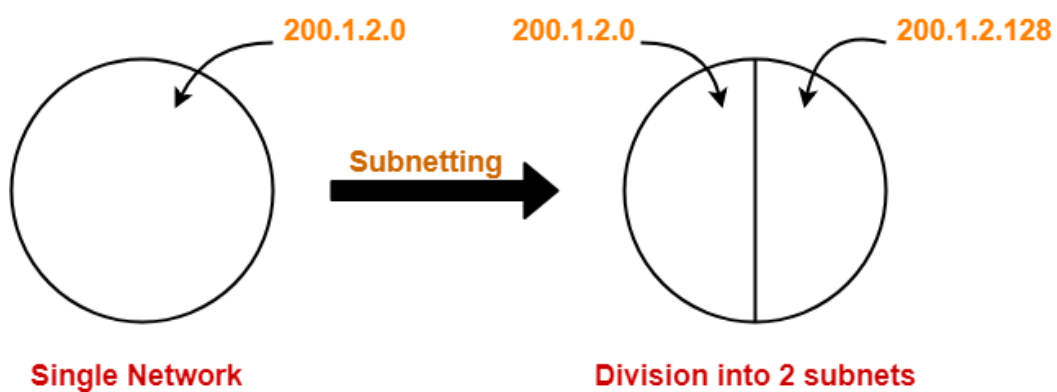
- Creating subnets:
 - borrow bits from the host ID
 - create a new Network Mask to show the new structure of the IPv4

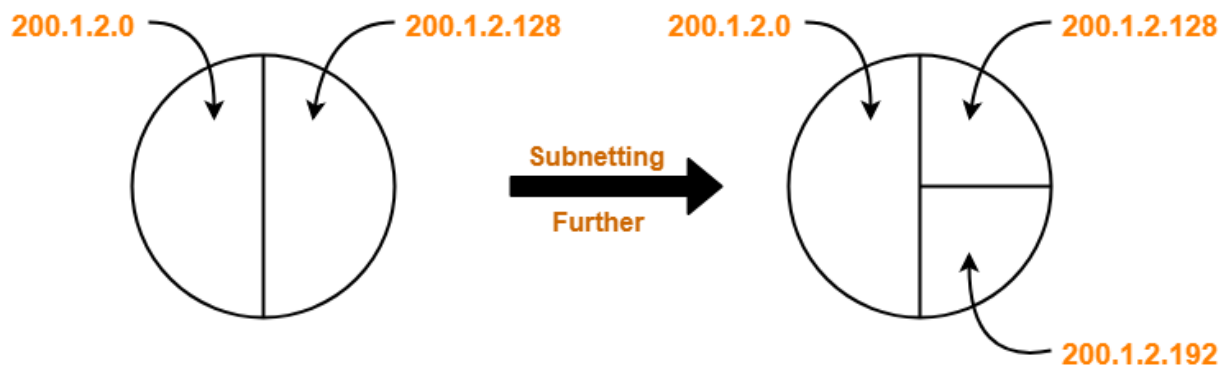
network ID	host ID
------------	---------

network ID	subnetwork ID	host ID
------------	---------------	---------



Subnetting a subnet ☺





Example: consider the network 192.168.4.0 /24. Divide this network into 7 subnets.

Step1: determine the number of bits needed

- how many bits do we need to create 7 subnets? 3 bits
- how many subnets can we compute? $2^3 = 8$ subnets (we will use 7 and 1 subnet we will keep it for future needs)

Step2: determine the new network mask (subnet mask)

- original NM: /24 = 255.255.255.0 or 11111111.11111111.11111111.00000000
- new NM: we need to borrow 3 bits for subnets (step 1) from the host part, so we mark this in the NM: 11111111.11111111.11111111.**111**00000 = /27 or 255.255.255.224

Step3: determine how many bits are remaining for the hosts

- how many bits are set to 0 in the new NM?
5 bits => 5 bits for hosts => so we can have $2^5 - 2 = 30$ hosts/subnet

Step4: compute IP subnets range

SN1:

Network bits	Subnet bits	Host bits	Conversion to decimal notation
11000000. 10101000.00000100. 000	000000		192.168.4.0 /27 - Network Address for SN1
11000000. 10101000.00000100. 000	00001		192.168.4.1 /27
11000000. 10101000.00000100. 000	00010		192.168.4.2 /27
.....			
11000000. 10101000.00000100. 000	11111		192.168.4.31 /27 - Broadcast Address for SN1

SN2:

Network bits	Subnet bits	Host bits	Conversion to decimal notation
11000000. 10101000.00000100. 001	00000		192.168.4.32 /27 - Network Address for SN2
11000000. 10101000.00000100. 001	00001		192.168.4.33 /27
11000000. 10101000.00000100. 001	00010		192.168.4.34 /27
.....			
11000000. 10101000.00000100. 001	11111		192.168.4.61 /27 - Broadcast Address for SN2

Compute the same for SN3 to SN8

	172	25	114	250		AND
IP Address (B class)	10101100	00011001	01110010	11111010		0 0 0
Network Mask	255	255	0	0	/16	0 1 0
Subnet Mask	255	255	255	192	/26	1 0 0
	11111111	11111111	11111111	11000000	AND	1 1 1
Subnet Address	10101100	00011001	01110010	11000000		
	172	25	114	192		
Subnet Broadcast	10101100	00011001	01110010	11111111		
	172	25	114	255		

Total number of host bits: 2^6	
Number of hosts: $2^6 - 2 = 64 - 2 = 62$	First host IP on subnet: 172.25.114.193
Total number of subnet bits: 2^{10}	Last host IP on subnet: 172.25.114.254
Number of subnets: $2^{10} = 1024$	

Exercises:

1. Determine the network and broadcast addresses and number of host bits and hosts for the given IPv4 addresses and prefixes:

IPv4 Address/Prefix	Network Address	Broadcast Address	Total Number of Host Bits	Total Number of Hosts
172.16.104.99/27				
198.133.219.250/24				
10.1.113.75/19				

2. Having the following information, compute subnets with the following constraints:

- A number of 62 subnets

Host IP Address: 172.16.0.0

Original Subnet Mask 255.255.0.0

3. Having the following information, compute subnets with the following constraints:

- A maximum number of 29 hosts/subnet

Host IP Address: 192.168.200.0

Original Subnet Mask 255.255.255.0

4. Having the following information, compute subnets with the following constraints:

- A number of 250 subnets

Host IP Address: 10.0.0.0

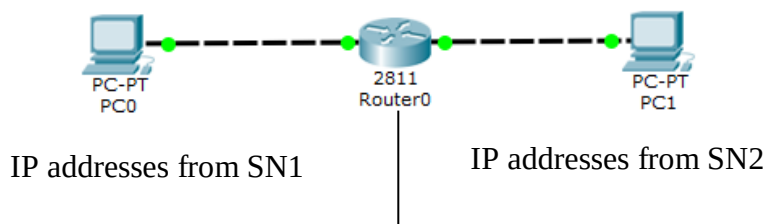
Original Subnet Mask 255.0.0.0

Test the connectivity on IPv4:

- **ipconfig /all** (**ifconfig** on Linux)
- **ipconfig /release** and **ipconfig /renew** (**dhclient** on Linux)
- **ping**, **tracert** (**traceroute** on Linux)
- **nslookup**
- **route print** (display routing table)

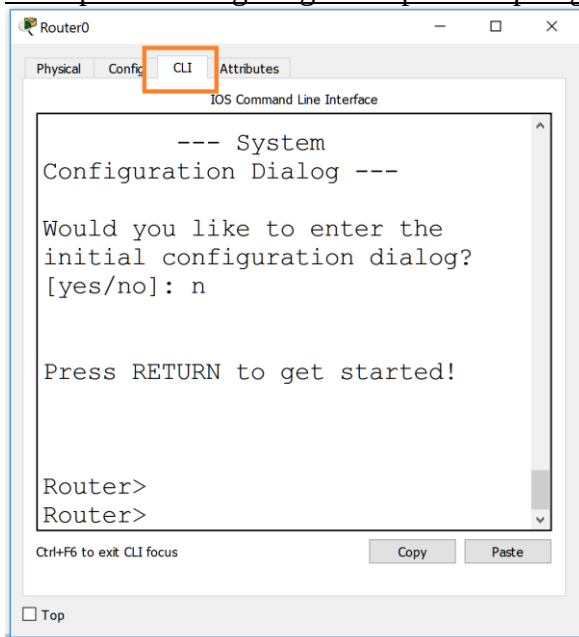
B. Packet Tracer

Laboratory test configuration:



Considering IP address 172.16.0.0 /16, compute 2 subnets and assign the correct IP address to the routers' interfaces and to the host computers (PC0 and PC1).

Example on configuring the depicted topology:



Step0: Enter configuration mode on the router

```
Router>enable
```

```
Router#configure terminal
```

```
Router(config)#
```

Step1: Assign static IPv4 address to the router interfaces

```
Router(config)#interface fastethernet 0/0
```

```
Router(config-if)#ip address 192.168.0.1 255.255.255.0
```

```
Router(config-if)#no shutdown
```

Description: Add an IP address on Fa0/0 interface

Step3: Display info

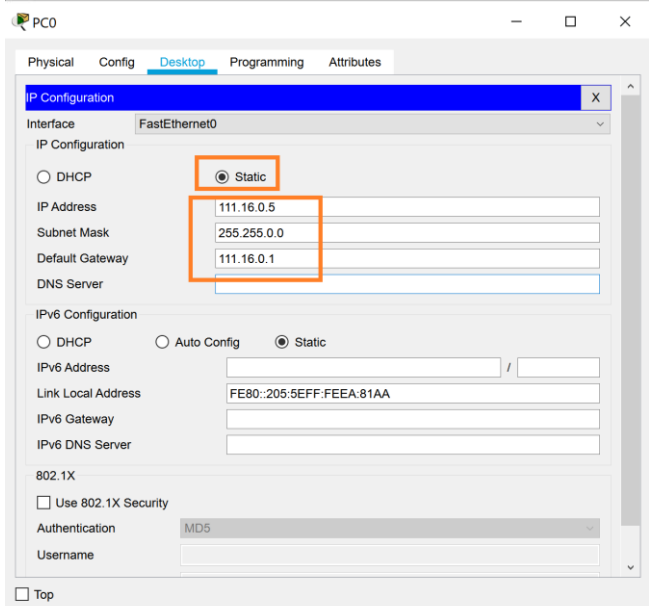
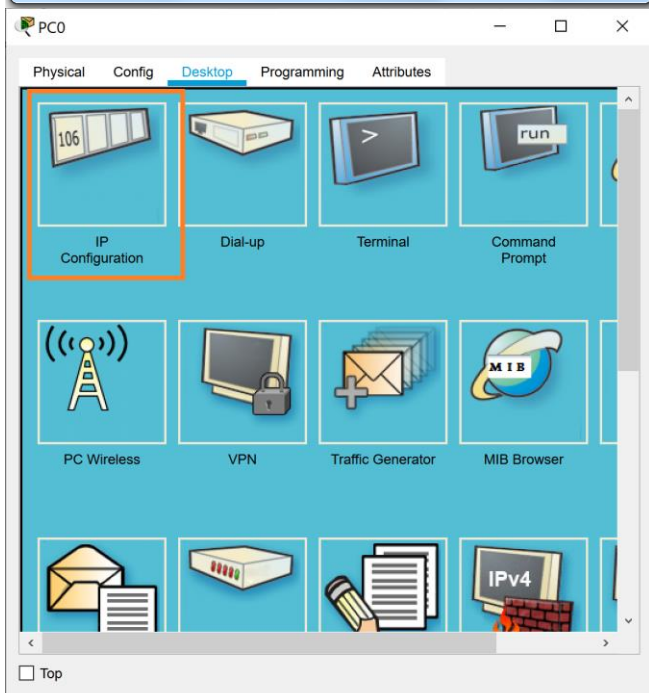
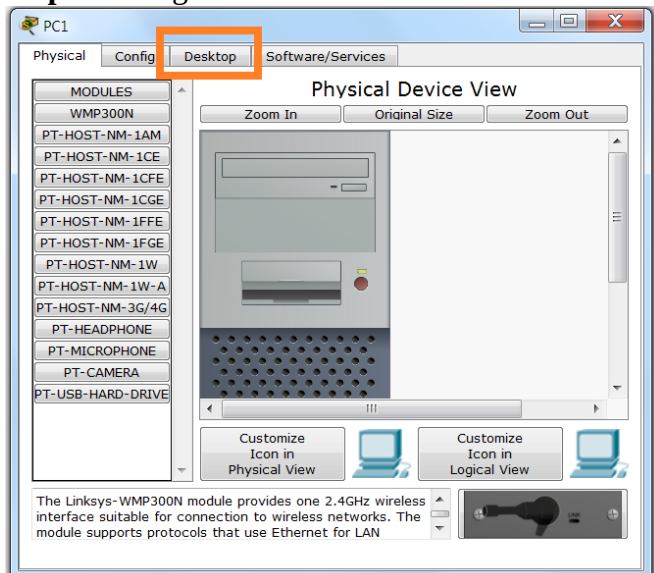
```
Router#show ip interface brief
```

Description: Display IP information about router's interfaces

```
Router#show ip route
```

Description: Display IP routing table

Step4: Configure IP addresses on the PCs



Test the connectivity.

a. check IP addresses oh hosts computers

b. ping <target IP>

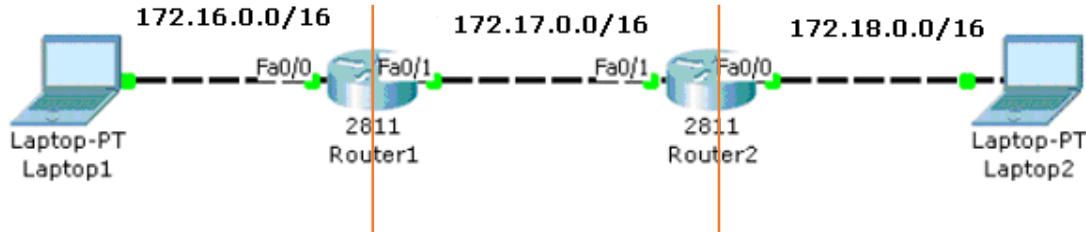
Lab - Guideline

Topics

- A. IPv4 routing
- B. DHCP IPv4

A. IPv4 routing

Laboratory test configuration:



Step1: Assign static IPv4 addresses to router interfaces and computers

Device	Interface	IP	NM	GW
Router1	Fa0/0	172.16.0.1	255.255.0.0	-
	Fa0/1	172.17.0.1	255.255.0.0	-
Router2	Fa0/0	172.18.0.1	255.255.0.0	-
	Fa0/1	172.17.0.2	255.255.0.0	-
Laptop1	Fa0	172.16.0.2	255.255.0.0	172.16.0.1
Laptop2	Fa0	172.18.0.2	255.255.0.0	172.18.0.1

Router1>enable

Router1#configure terminal

Router1(config)#**interface fastethernet 0/0**

Router1(config-if)#ip address 172.16.0.1 255.255.0.0

Router1(config-if)#no shutdown

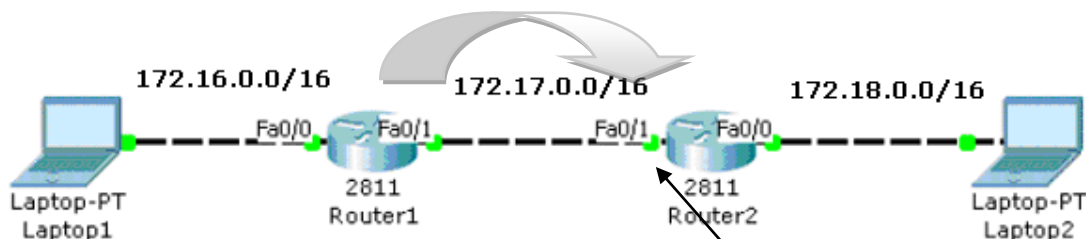
Router1(config-if)#exit

Description: Add an IP address on Fa0/0 interface

Step2: Set static routes

General syntax:

Router(config)#ip route netw_dest_address next_hop_address/interface



Router1(config)#ip route 172.18.0.0 255.255.0.0 172.17.0.2

Destination Network (Network Address)

Next_Hop Address

Router1#show ip route

Description: Visualize the routing table

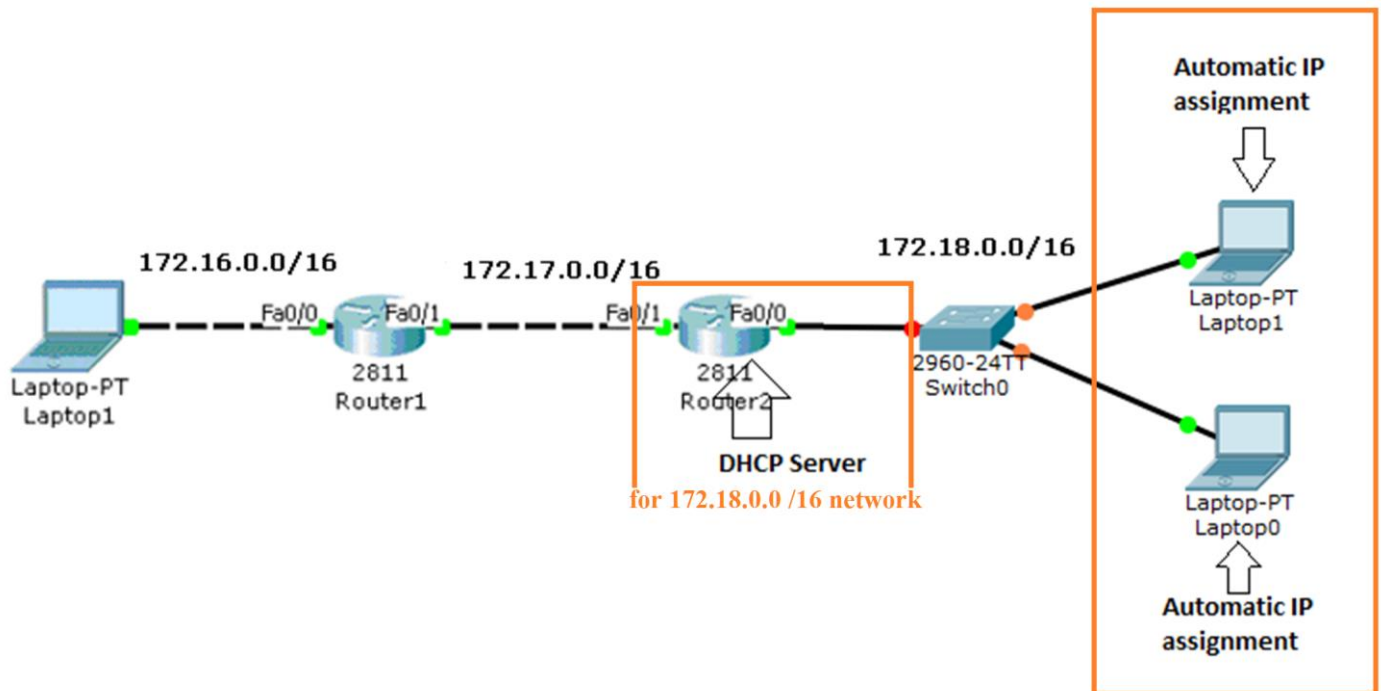
Test the connectivity.

- a. ping <target IP>
- b. tracert <target IP>

B. DHCP

A Dynamic Host Configuration Protocol (DHCP) server assigns and manages IP addresses from specified address pools within the device to DHCP clients.

Laboratory test configuration:



Step: Configure DHCP (commands)

Configure the pool name

```
Router(config)# ip dhcp pool name
```

Configure the addresses

```
Router(dhcp-config)# network network-number [mask|prefix-length]
```

Setting up the IP Domain Name System Servers for the Client

```
Router(dhcp-config)# dns-server address [address2 ...address5]
```

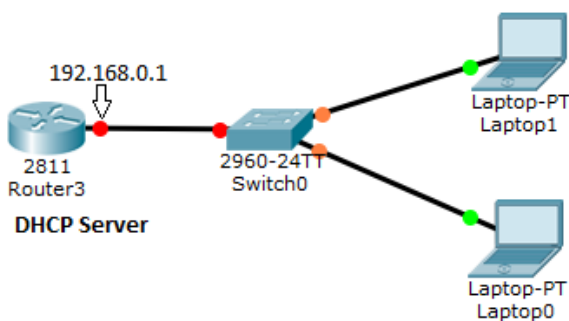
Configuring the Default Router for the Client (!the default router has to be on the same subnet as the DHCP client)

```
Router(dhcp-config)# default-router address
```

Exclude IP addresses

```
Router(config)#ip dhcp excluded-address start_address end_address  
(Exclude IP addresses that were statically assigned)
```

Example:



```
Router(config)# ip dhcp pool MY_DHCP  
Router(dhcp-config)# network 192.168.0.0 255.255.255.0  
Router(dhcp-config)# dns-server 8.8.8.8  
Router(dhcp-config)# default-router 192.168.0.1  
Router(dhcp-config)# exit  
Router(config)#ip dhcp excluded-address 192.168.0.1
```

Test the connectivity.

- a. *check IP addresses oh hosts computers*
- b. *ping <target IP>*

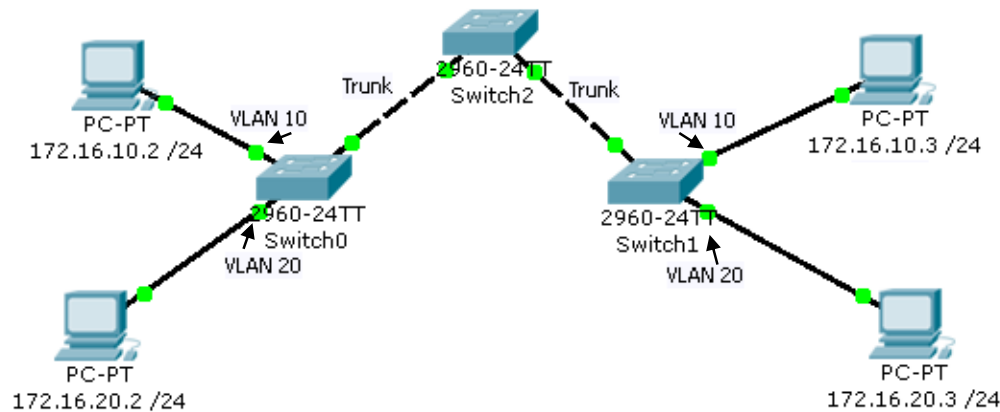
Lab Guideline

Topics

- A. VLANs and Trunking: IEEE 802.1Q
- B. Inter-VLAN routing

A. VLANs and Trunking

Laboratory test configuration:



IMPORTANT! Create VLAN 10 and 20 on all switches!

Commands Used:

Switch>enable

Switch#configure terminal

Switch(config)#vlan 10

Description: Global configuration command that creates VLAN 10

Switch(config-vlan)#name Student_Vlan

Description: Assigns a name to the VLAN

Switch#show vlan brief

Description: Displays VLANs information (the contents of the vlan.dat file)

Switch(config)#interface fa 0/1

Switch(config-if)#switchport mode access

Switch(config-if)#switchport access vlan 10

Description: Configures a port as an access port and assigns VLAN 10 to an access port

Switch(config)#interface fa 0/2

Switch(config-if)#switchport mode trunk

Description: Configures a port as a trunk port

Switch#show interfaces trunk

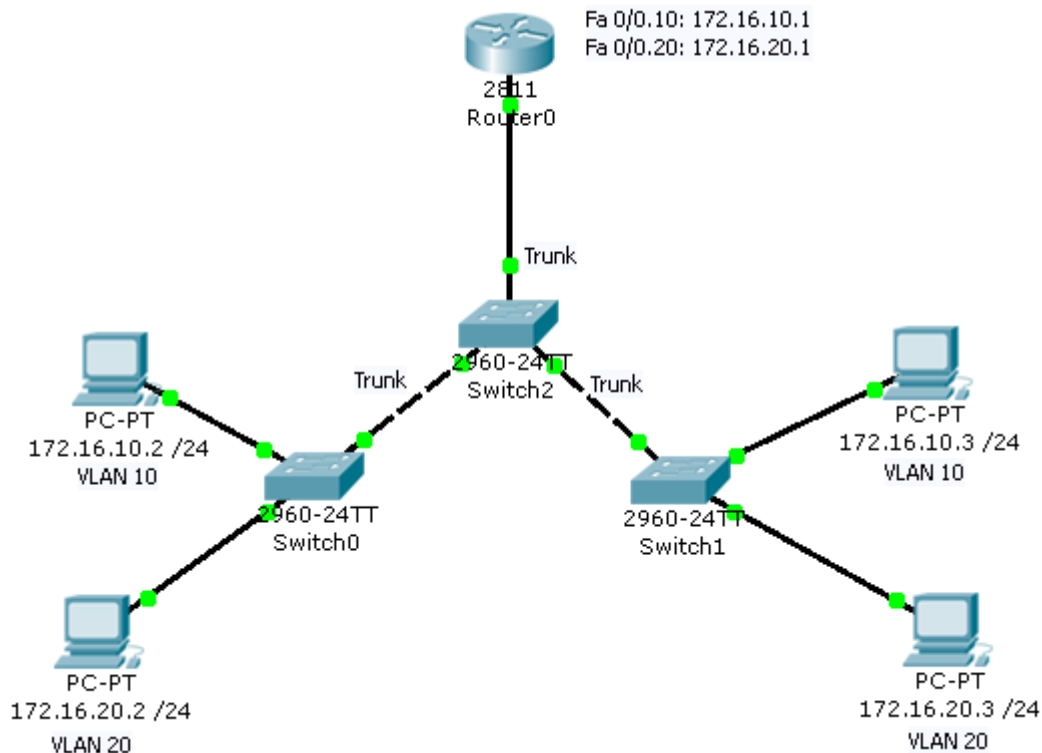
Description: Displays trunking information for the switch

Repeat the above steps for the other interfaces on the corresponding switches

Test the connectivity.

B. Inter-VLAN routing

Laboratory test configuration:



Commands Used:

Router(config)#interface fastethernet 0/0

Router(config-if)#no shutdown

Description: Enable Fa0/0 interface

Router(config)#interface fastethernet 0/0.10

Description: Creates a subinterface numbered 10 on the router for Fa0/0

Router(config-subif)#encapsulation dot1q 10

Description: Specifies IEEE 802.1Q as the VLAN tagging method for VLAN 10 on this subinterface

Router(config-subif)#ip address 172.16.10.1 255.255.255.0

Description: Add an IP address on Fa0/0.10 subinterface

Repeat the above steps to create the second subinterface (fastEthernet 0/0.20) on the router

Also, on Switch2, configure the interface connecting the router in Trunk mode. Why is that necessary?

Router#show ip route

Description: Visualize the routing table

Add Gateways to PCs - the IP addresses set on the Multilayer Switch for the respective VLAN:

- 172.16.10.1 for devices in VLAN 10
- 172.16.20.1 for devices in VLAN 20

Test the connectivity.

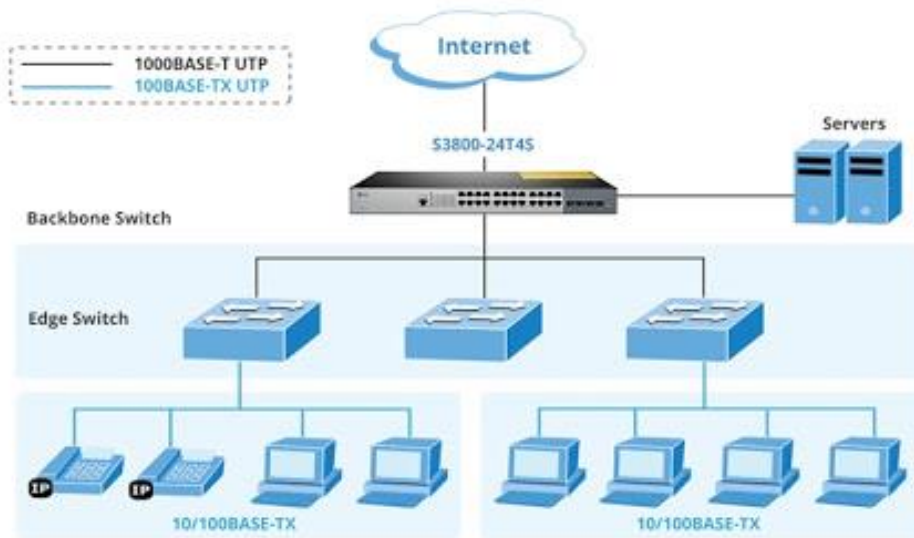
ping <IP>

tracert <IP>

Lab - Guideline

Topics

- A. Switches and bridges
- B. Spanning-Tree Protocol



A. Switches and bridges

- layer 2 devices
- switching data frames and maintaining switching operations

Switching is classified as

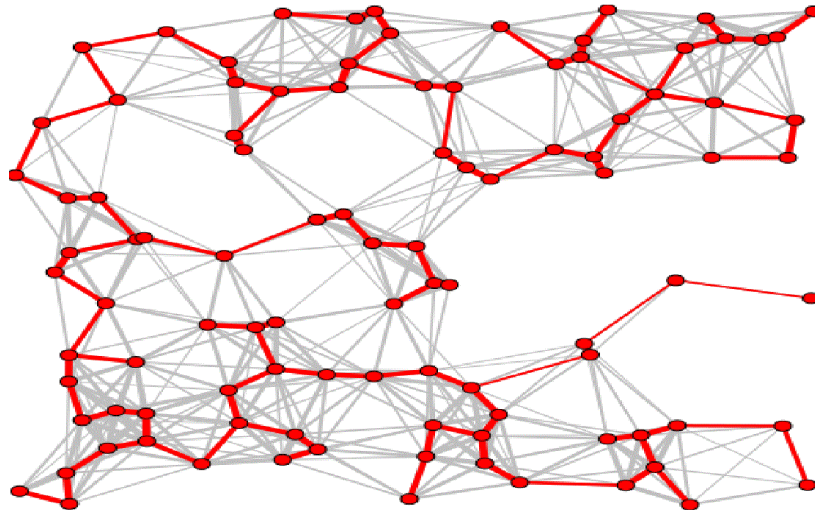
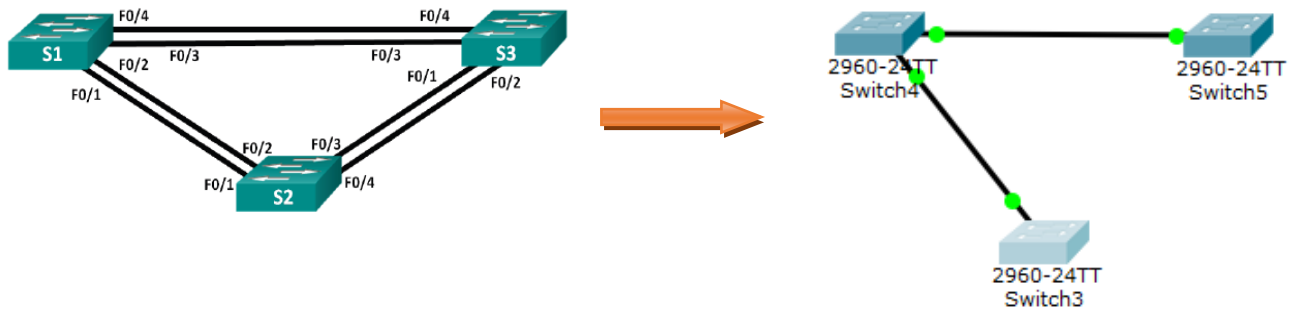
- symmetric
- asymmetric

Switching modes are classified as:

- store-and-forward: buffer entire frame
- cut-through: read at least destination address (MAC)
 - o fast-forward: read destination address (MAC)
 - o fragment-free: read the first 64 bytes (512 bits) of the frame

Ethernet II Frame Structure and Field Size					
8 Bytes	6 Bytes	6 Bytes	2 Bytes	46 - 1500 Bytes	4 Bytes
Preamble	Destination Address	Source Address	Type	Data	Frame Check Sequence

B. Spanning-Tree Protocol



BPDU – every 2 secs:

Root BID	Root Path Cost	Sender BID	Port ID
----------	----------------	------------	---------

BPDU structure

Bridge Priority	MAC address
-----------------	-------------

BID structure

Port Priority	Port Number
---------------	-------------

Port ID structure

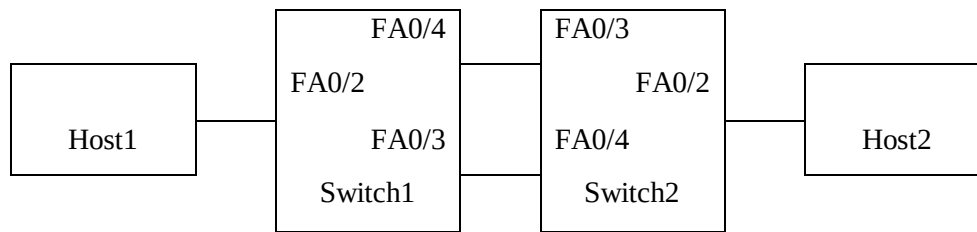
Link Speed	Cost
10Mbps	2000000
100Mbps	200000
1Gbps	20000
10Gbps	2000
1Tbps	20
10Tbps	2

Link Cost/Speed

Elect root bridge -> calculate the shortest path from itself to the root bridge --> Spanning-Tree

Port states: blocking, listening, learning, forwarding and disabled

Lab configuration



Choose IP addresses for the hosts and the switches (from the same network – example 192.168.1.0/24)

Switch(config)# hostname *host_name*

- Replace *host_name* with your name

Configure the primary terminal line with the following commands:

```
# line console 0
# password password
# login
# exit
```

- Replace *password* with your name

Configure virtual terminal with the following commands:

```
# line vty 0 4
# password password
# login
# exit
```

- Replace *password* with your name

Configure IP address and default gateway with the following commands:

```
# interface VLAN1
# ip address ip_address netmask
# no shutdown
# exit
# ip default-gateway default_gateway_address
```

Visualize spanning-tree configuration

Switch# show spanning-tree

Change switch priority

Switch(config)# spanning-tree vlan 1 priority 0

Change switch port priority

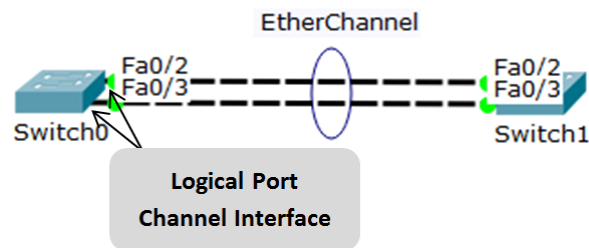
Switch(config)# interface fastethernet 0/4

Switch(config-if)# spanning-tree vlan 1 port-priority 0

Visualize spanning-tree configuration

Switch# show spanning-tree

EtherChannel creates an aggregation that is seen as one logical link => redundancy



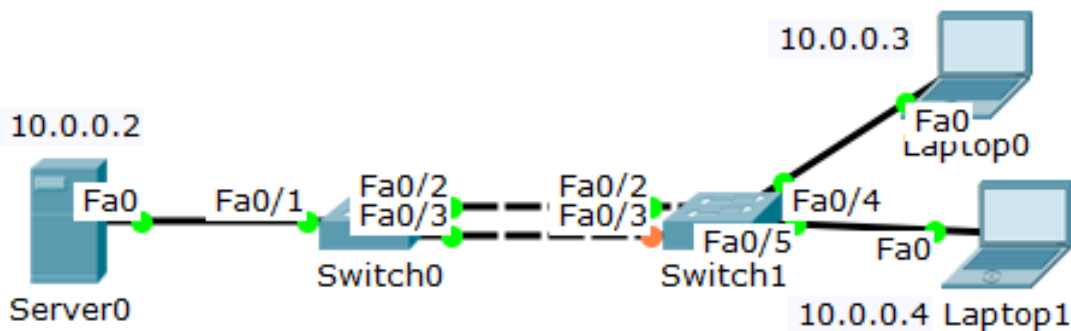
Usually each EtherChannel can consist of up to 8 compatibly-configured Ethernet ports.

Cisco IOS switch can usually support 6 EtherChannels.

- all physical links in the EtherChannel are active because STP sees only one logical link.
- the loss of one physical link within the channel does not create a change in the topology => a spanning-tree recalculation is not required
- the spanning-tree cost is calculated based on the number of ports assigned to the port-channel and it does not dynamically change when links go down or are brought back up within the port-channel
- If the physical ports of one side are configured as trunks, the physical ports of the other side must also be configured as trunks within the same native VLAN

Protocols:

1. Port Aggregation Protocol (PAgP)
2. Link Aggregation Control Protocol (LACP)



Examine and explain the output of this command.

```
Switch0#show spanning-tree
```

```
Switch0(config)#interface range fastEthernet 0/2-3
```

Specify the interfaces that compose the EtherChannel group

```
Switch0(config-if-range)#channel-group 1 mode on
```

Create the port channel interface 1

```
Switch0#show etherchannel summary
```

View the Etherchannel information with the

```
Switch0(config)#interface port-channel 1
```

```
Switch0(config-if)#switchport mode trunk
```

Configure the EtherChannel as a trunk interface

```
Switch0(config)#port-channel load-balance dst-mac
```

Configure EtherChannel load balancing method

```
Switch0#show etherchannel load-balance
```

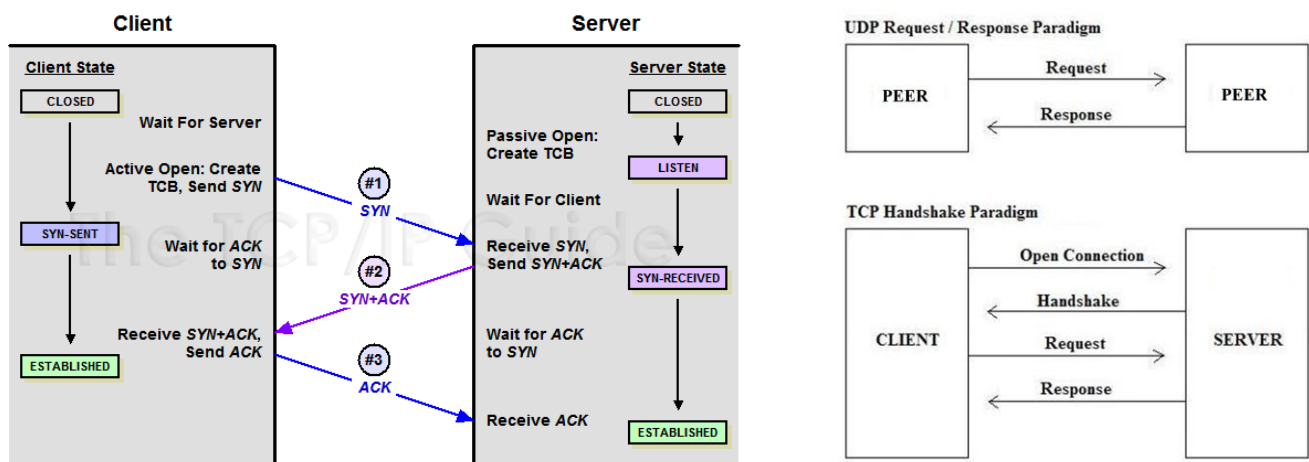
View the EtherChannel load balancing method information

Lab - Guideline

A. TCP and UDP

- Description for both protocols

- **UDP - User Datagram Protocol**
 - Connectionless
 - support for the DNS and TFTP protocols
- **TCP - Transmission Control Protocol**
 - Connection-oriented
 - 3-way handshake
 - Closing: FIN signal
 - support for the HTTP and FTP protocols

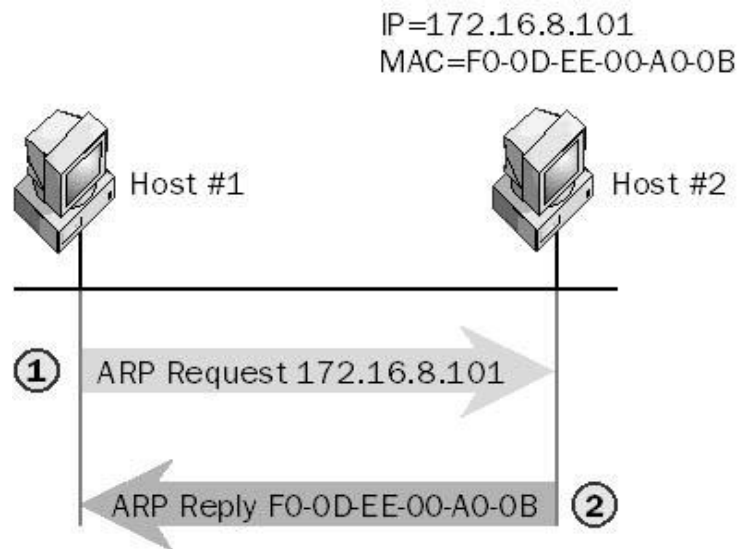


- **Socket** definition: **(IP+Port)** +Protocol (TCP/UDP)

B. Wireshark

Packet capturing and protocols analysis with **Wireshark**, on several protocols:

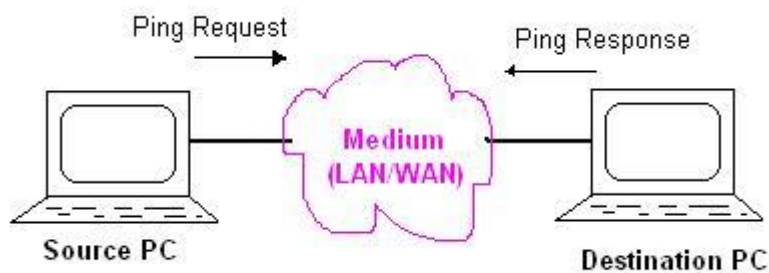
- **ARP** and Ethernet protocols



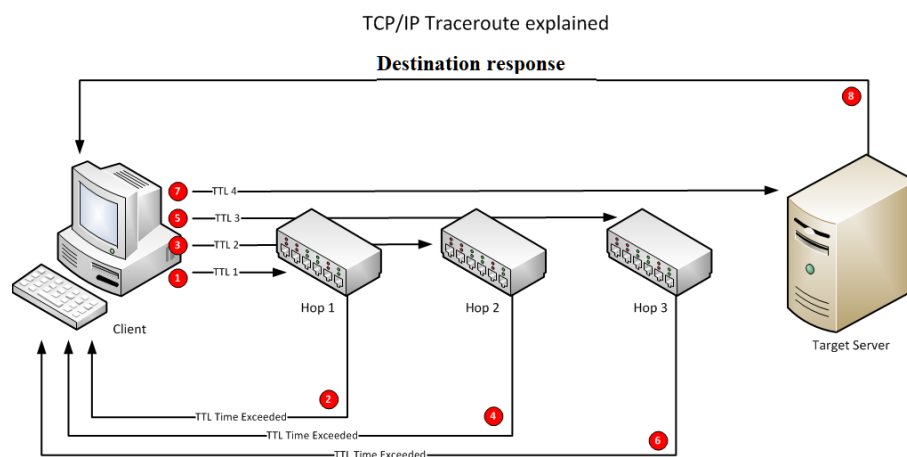
- IPv4 and IPv6 protocols

- ICMP protocol

ping



tracert / pathping



- DNS protocol

nslookup

UDP protocol + port (53 for DNS)

```

Command Prompt - nslookup

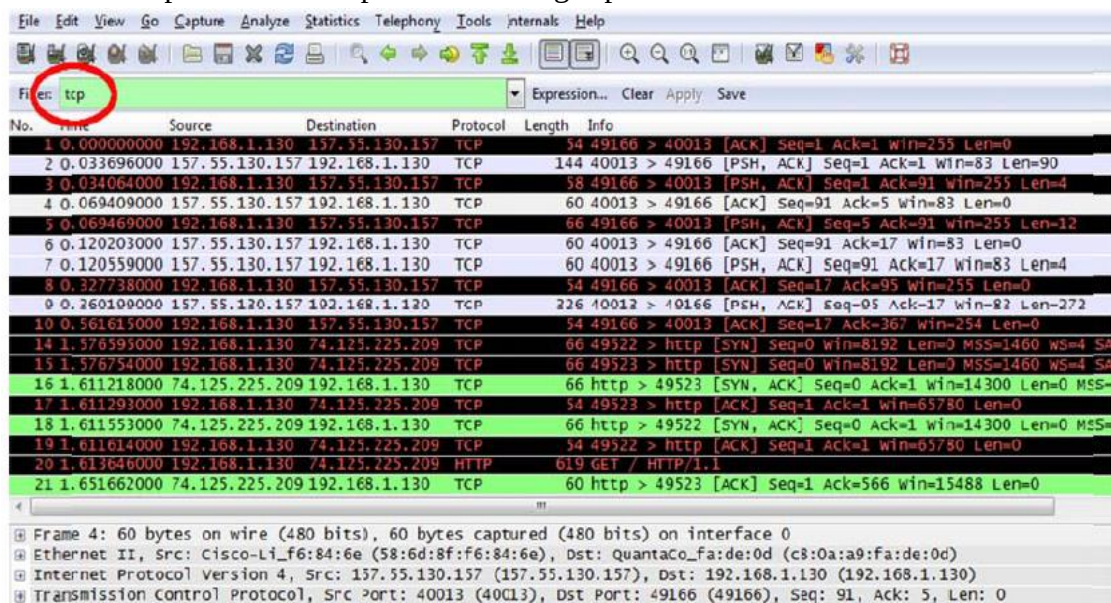
C:\Users\B>nslookup
Default Server:  rdns.utcluj.ro
Address:  193.226.5.151

> www.utcluj.ro
Server:  rdns.utcluj.ro
Address:  193.226.5.151

Non-authoritative answer:
Name:     www.utcluj.ro
Addresses: 2001:b30:5000:2::7
          193.226.5.7
  
```

- HTTP and FTP protocols

TCP protocol and triple handshaking + ports



The image shows a Wireshark packet capture interface. The 'Filter' field is set to 'tcp' and is circled in red. The packet list shows a series of TCP packets (1-21) and one HTTP packet (20). The packet details pane shows the structure of the selected packet (Frame 20: 619 bytes on wire).

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000000	192.168.1.130	157.55.130.157	TCP	54	49166 > 40013 [ACK] Seq=1 Ack=1 win=255 Len=0
2	0.033696000	157.55.130.157	192.168.1.130	TCP	144	40013 > 49166 [PSH, ACK] Seq=1 Ack=1 win=83 Len=90
3	0.034064000	192.168.1.130	157.55.130.157	TCP	58	49166 > 40013 [PSH, ACK] Seq=1 Ack=91 win=255 Len=4
4	0.069409000	157.55.130.157	192.168.1.130	TCP	60	40013 > 49166 [ACK] Seq=91 Ack=5 win=83 Len=0
5	0.069469000	192.168.1.130	157.55.130.157	TCP	66	49166 > 40013 [PSH, ACK] Seq=5 Ack=91 win=255 Len=12
6	0.120203000	157.55.130.157	192.168.1.130	TCP	60	40013 > 49166 [ACK] Seq=91 Ack=17 win=83 Len=0
7	0.120559000	157.55.130.157	192.168.1.130	TCP	60	40013 > 49166 [PSH, ACK] Seq=91 Ack=17 win=83 Len=4
8	0.327738000	192.168.1.130	157.55.130.157	TCP	54	49166 > 40013 [ACK] Seq=17 Ack=95 win=255 Len=0
9	0.260100000	157.55.120.157	102.168.1.120	TCP	226	40012 > 40166 [PSH, ACK] Seq=95 Ack=17 win=82 Len=272
10	0.561615000	192.168.1.130	157.55.130.157	TCP	54	49166 > 40013 [ACK] Seq=17 Ack=367 win=254 Len=0
14	1.576595000	192.168.1.130	74.125.225.209	TCP	66	49522 > http [SYN] Seq=0 win=8192 Len=0 MSS=1460 WS=4 SA
15	1.576754000	192.168.1.130	74.125.225.209	TCP	66	49523 > http [SYN] Seq=0 win=8192 Len=0 MSS=1460 WS=4 SA
16	1.611218000	74.125.225.209	192.168.1.130	TCP	66	http > 49523 [SYN, ACK] Seq=0 Ack=1 win=14300 Len=0 MSS=
17	1.611293000	192.168.1.130	74.125.225.209	TCP	54	49523 > http [ACK] Seq=1 Ack=1 win=65780 Len=0
18	1.611553000	74.125.225.209	192.168.1.130	TCP	66	http > 49522 [SYN, ACK] Seq=0 Ack=1 win=14300 Len=0 MSS=
19	1.611614000	192.168.1.130	74.125.225.209	TCP	54	49522 > http [ACK] Seq=1 Ack=1 win=65780 Len=0
20	1.613646000	192.168.1.130	74.125.225.209	HTTP	619	GET / HTTP/1.1
21	1.651662000	74.125.225.209	192.168.1.130	TCP	60	http > 49523 [ACK] Seq=1 Ack=566 win=15488 Len=0

Frame 4: 60 bytes on wire (480 bits), 60 bytes captured (480 bits) on interface 0
 Ethernet II, Src: Cisco-Li_f6:84:6e (58:6d:8f:f6:84:6e), Dst: Quantaco_fa:de:0d (c8:0a:a9:fa:de:0d)
 Internet Protocol Version 4, Src: 157.55.130.157 (157.55.130.157), Dst: 192.168.1.130 (192.168.1.130)
 Transmission Control Protocol, Src Port: 40013 (40013), Dst Port: 49166 (49166), Seq: 91, Ack: 5, Len: 0

- Wireshark statistics